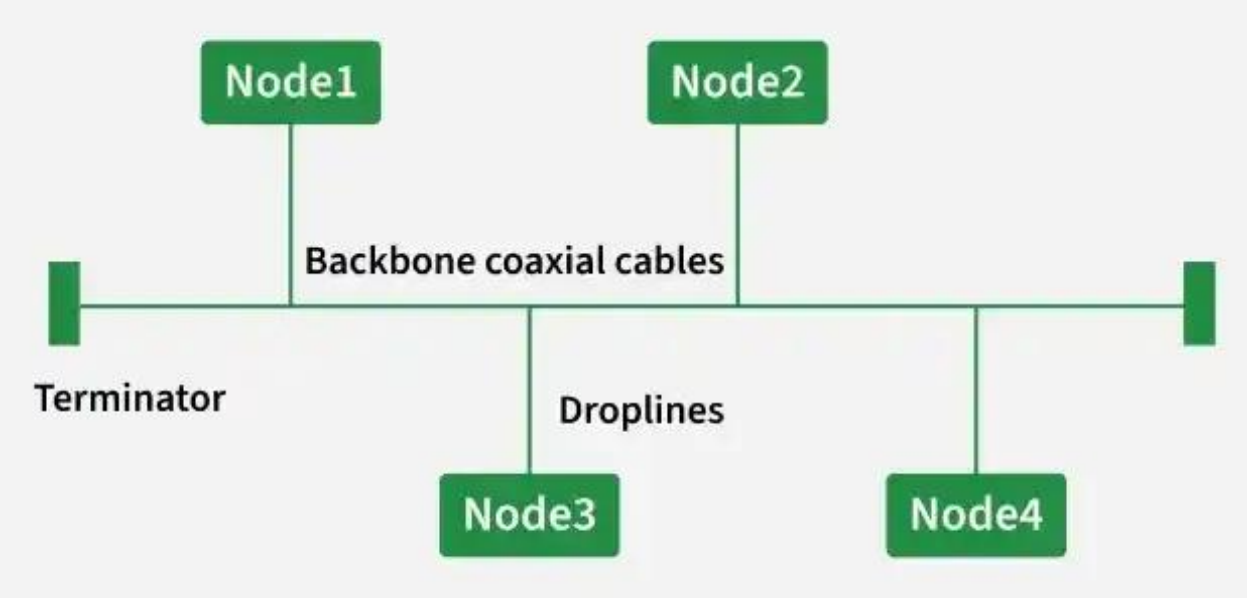
Day 2: Network TopologiesTheory

Task 1 : Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.



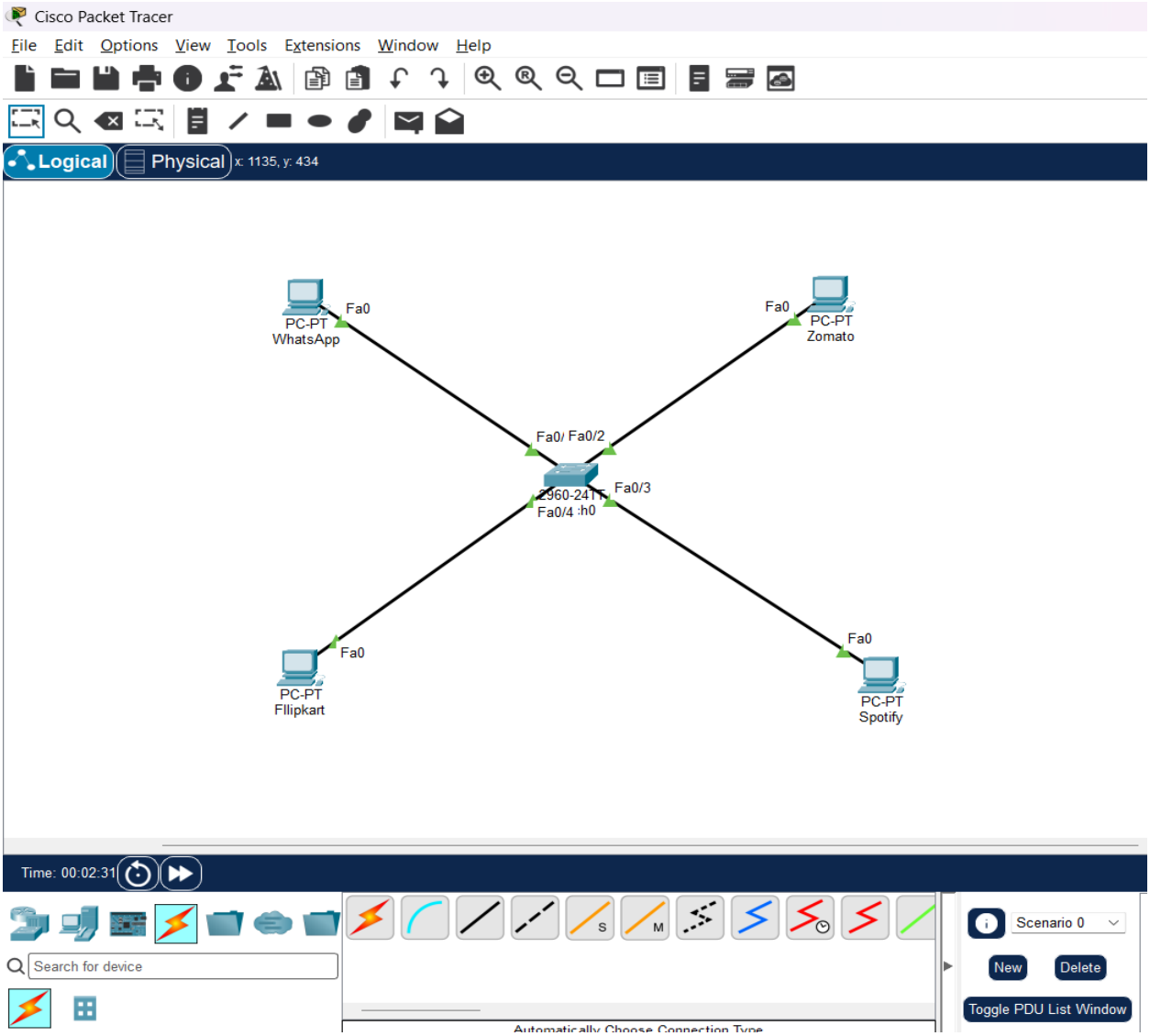
Key Components

1. **Computers/Nodes:** Devices connected to the shared communication cable.
2. **Backbone Cable:** The main cable that carries network signals.
3. **Drop Lines:** Connections between the computers and the backbone cable.
4. **Terminators:** Devices at both ends of the cable that prevent signal reflections.

Real-World Example

Older Ethernet networks using 10BASE2 coaxial cable used a bus topology to connect multiple computers through a shared cable.

Task 2 : In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.



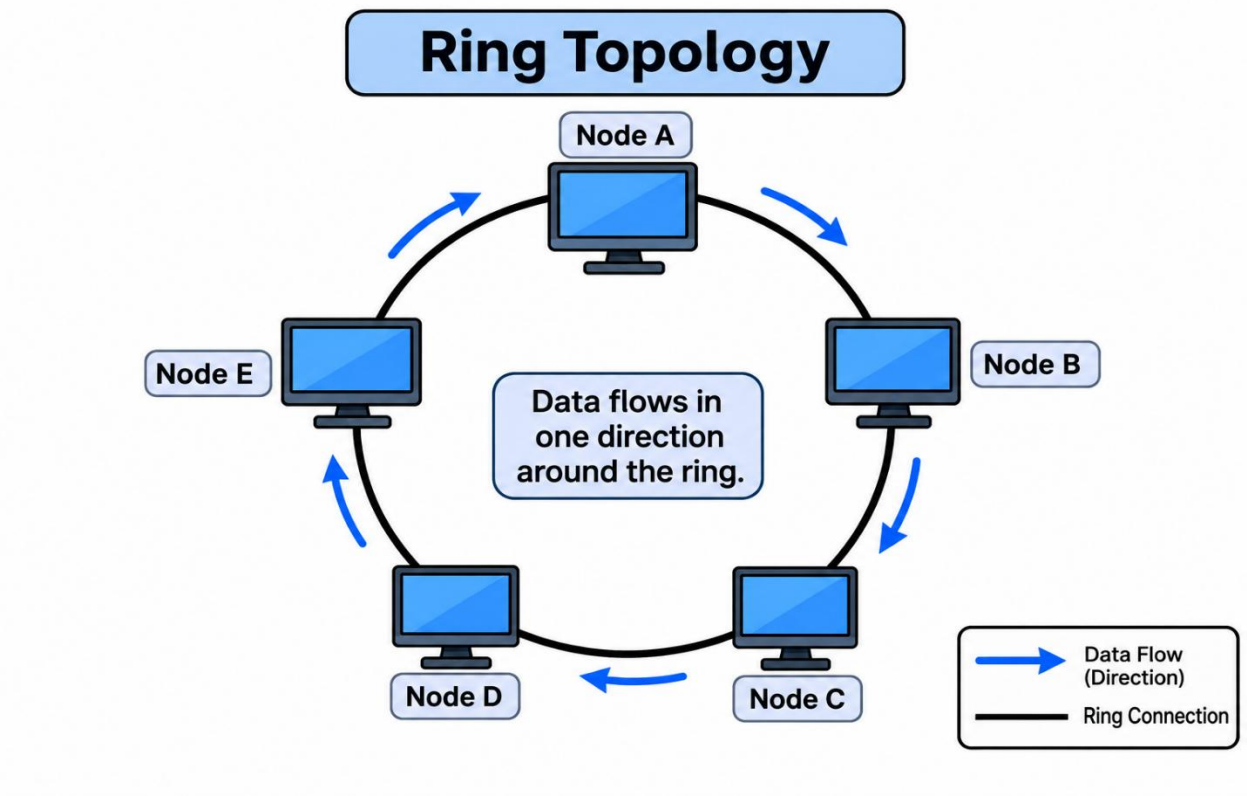
Devices Required

- 1 Cisco 2960 switch
- 4 PCs
- 4 Copper Straight-Through cables

Result

All four PCs should be connected to the central switch, forming a star topology.

Task 3 : Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.



Diagram

A ring topology connects each device to two neighboring devices, forming a closed loop. Arrows indicate the direction in which data travels around the ring.

How Data Travels

In a ring topology, data travels from one device to the next around the ring until it reaches its destination. In a unidirectional ring, data moves in one direction only. Some ring technologies use special mechanisms to manage access to the network and reduce data collisions.

Conclusion

Ring topology provides an organized path for data transmission, but a break in a basic ring can disrupt communication unless a protection or recovery mechanism is available.

Task 4 : Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Topology	Advantages	Disadvantages
Mesh	1. Multiple paths can provide high reliability and fault tolerance. 2. A link failure may not interrupt communication if an alternative path exists.	1. Requires many cables and ports, increasing installation costs. 2. Complex to install, manage, and maintain, especially in a full mesh.
Star	1. Easy to install, manage, and troubleshoot. 2. Failure of one device's cable usually does not affect other devices.	1. Failure of the central switch can disrupt the entire star network. 2. Performance depends on the capacity of the central switch and its links.